Gate Topology and Reproducibility of Prior-Guided Spectral Gating Across Near-Infrared and Raman Spectroscopy

Guilherme Coelho[a], Clarimar José Coelho[b,*]

[a]*State University of Campinas, Cidade Universitária Zeferino Vaz - Barão Geraldo, Campinas, 13083–970, São Paulo, Brazil*
[b]*Postgraduate Program in Industrial Engineering and Artificial Intelligence Pontificia Universidade Catolica de Goias, Primeira Avenida, s/n - Quadra 88 - Setor Leste Universitário, Goiânia, 74605–010, Goiás, Brazil*

Abstract

Prior-Guided Spectral Gating (PGSG) is a spectral attention mechanism in which a learnable per-variable sigmoid gate, initialized from a literature-derived prior, weights the spectrum before a shallow multilayer perceptron. It has so far been studied only on near-infrared (NIR) spectroscopy. We apply the unmodified architecture to public NIR (n = 1,448, mango dry matter) and Raman (n = 6,472, glucose) benchmarks and ask three questions: whether the model outperforms partial least squares (PLS), how the learned gate topology differs between modalities, and what a literature-informed prior contributes. Against PLS with the number of latent variables selected by cross-validation (one-standard-error rule), and with train/test splits that keep replicate spectra of the same sample together, the model does not outperform PLS in NIR (paired $\Delta R^2 = -0.108 \pm 0.031$ over ten splits, lower in all ten) but does in Raman ($\Delta R^2 = +0.036 \pm 0.007$, higher in all ten). The NIR advantage reported with a fixed, under-parameterized PLS baseline (+0.23) disappears once the baseline is calibrated. The learned gates nevertheless differ between modalities in a physically consistent way: Raman gates are sparser than NIR gates, and NIR gates are smoother once smoothness is normalized by physical band density, a correction we show to be necessary. A literature-informed prior does not change accuracy but substantially in-

*Corresponding author.
Email address: clarimar@pucgoias.edu.br (Clarimar José Coelho)

creases gate reproducibility across training seeds (pairwise Jaccard similarity 0.90–0.94 versus 0.60–0.65). A band-level test relating local gate structure to vibrational linewidths was negative. Since an uninformed initialization reaches the same accuracy, the predictive gain in Raman belongs to the nonlinear regressor rather than to the prior, whose value lies in interpretive stability. Predictive claims for neural spectral models require a baseline whose complexity is selected from the data and a validation design that respects replicate structure.

Keywords: spectral gating, interpretable deep learning, Raman spectroscopy, near-infrared spectroscopy, chemometrics, cross-modality generalization

---

1. Introduction

Prior-Guided Spectral Gating (PGSG) is a chemometrically-informed attention mechanism for spectroscopic regression, in which a learnable gate $g \in (0,1)^p$ over p spectral variables is initialized from a literature-derived prior s, rather than from the training data, and multiplies the input spectrum before a downstream regressor. The mechanism was originally established on near-infrared (NIR) spectroscopy at small sample sizes [1]. The present study uses a revised formulation of the architecture, developed within the same research program and fully specified in Section 4.3: the gate of the original formulation, a temperature-scaled softmax normalized over all spectral variables, is replaced by an independent sigmoid gate per variable, hereafter `PGSGv2Model`. Both formulations feed a compact MLP and are trained end to end by automatic differentiation; the change removes the coupling that a normalized softmax imposes between the weights of different variables, under which raising one gate necessarily lowers the others.

PGSG has so far been evaluated exclusively on NIR, leaving open a basic question about the generality of the mechanism: does prior-guided gating help because of a property specific to NIR spectra, or does it reflect a principle applicable across spectroscopic modalities with different physical structure? NIR spectra are dominated by broad, overlapping overtone and combination bands, in which analyte-relevant signal is typically diluted across many correlated wavelengths [2]. Raman spectra, by contrast, exhibit narrow, well-resolved vibrational peaks superimposed on a broad fluorescence background, which becomes approximately flat only after baseline correction [3]. If the

value of prior-guided gating in NIR derives from down-weighting broad, uninformative regions in favor of a chemically relevant subset, the mechanism might contribute comparatively little in a modality where informative peaks are already narrow and readily isolated by the regressor.

We test this directly by applying the unmodified `PGSGv2Model` architecture to a Raman regression task, by comparing it against a PLS baseline whose number of latent variables is selected by cross-validation, and by formalizing quantitative descriptors of learned-gate topology — entropy, smoothness, and sparsity — to characterize *how*, not merely *whether*, gate structure differs between the two modalities. We evaluate five falsifiable hypotheses (Section 3) end to end on public NIR and Raman benchmarks and, in the interest of transparency, document two corrections made during this study: a model-identity error during execution, and a fixed-complexity PLS baseline in the submitted version whose calibration reversed the conclusion for H1 (Section 5).

2. Scope

An early planning stage of this work considered simultaneous generalization to four modalities: NIR, Raman, mid-infrared (MIR), and mass spectrometry (MS). This scope was narrowed to two modalities for two reasons. First, validating across four modalities within a single study would create an empirical scope exceeding what can be executed and tested with rigor. Second, the physical argument motivating spectral gating — correlation between adjacent positions along a continuous spectral axis — does not hold for MS, whose signals are discrete mass-to-charge values without an analogous local neighborhood structure; including MS without adapting the gating operator itself would constitute a weak extension. MIR and MS therefore remain outside the empirical scope of this study.

3. Hypotheses

H1 (generalization without architectural change). `PGSGv2Model`, applied to Raman data without architectural modification (only reconfiguration of the initialization prior), outperforms PLS, reproducing the NIR result qualitatively ($\Delta R^2_{\text{PLS}} > 0$).

H2 (differential sparsity). The gate learned on Raman data is significantly more sparse (Hoyer index) than the gate learned on NIR data,

reflecting the narrow-peak structure of Raman scattering relative to the broad, overlapping overtone and combination bands of NIR.

H3 (differential smoothness). The gate learned on NIR data is significantly smoother (normalized total variation) than on Raman.

H4 (value of a physically motivated prior). Initializing the gate from literature-derived vibrational assignments, rather than from an uninformed initialization, yields predictive performance that is equal to or better, and gate stability across training seeds that is greater, in both modalities.

H5 (structure–physics correlation). The entropy and smoothness of the learned gate correlate with the known spectral bandwidth of the vibrational transitions relevant to the target analyte, within a given modality.

4. Materials and Methods

4.1. Datasets

The NIR branch reuses the Mango DMC v3 dataset (dry-matter content, %; n = 1,448 spectra from Season 4) [4], with a literature-derived prior [5] (Section 4.4). The Raman branch uses `bioprocess_substrates` [6], a bioprocess-monitoring benchmark accessed through the RamanBench collection [7], with glucose concentration as the regression target. Of 6,960 spectra in the source dataset, 6,472 carry a valid glucose label and were retained (p = 1,870 Raman-shift variables spanning 390.6–3384.7 cm^{-1}). Both datasets contain replicate spectra of the same physical sample: the 1,448 NIR spectra come from 725 fruits (723 fruits measured twice, with identical metadata and dry-matter value), and the 6,472 Raman spectra from 3,191 distinct mixtures (identical concentration vectors; mostly duplicate acquisitions). Table 1 summarizes both branches.

This dataset was selected, among 58 regression datasets available in RamanBench, on the basis of three criteria: a real (non-simulated) measurement, a single chemically interpretable target, and a sample size in the thousands, comparable to that of the NIR branch (n = 6,472 Raman versus n = 1,448 NIR). The latter criterion avoids the small-sample regime that characterizes most publicly available Raman regression datasets and that would otherwise confound modality with sample size; alternative candidates

considered and not selected for this reason include organic-acid concentration datasets from inline process monitoring [8], which are real measurements but of substantially smaller sample size.

Table 1: Summary of the two experimental branches.

	NIR	Raman
Dataset	Mango DMC v3, Season 4	`bioprocess_substrates`
Target	Dry matter content (%)	Glucose concentration
n (valid spectra)	1,448	6,472
Physical samples	725 fruits	3,191 mixtures
p (variables, after cleaning)	281	1,870
Spectral range	285–1200 nm	390.6–3384.7 cm^{-1}
Train/test split (H1)	80/20, grouped by fruit	80/20, grouped by mixture

4.2. Preprocessing

The NIR branch reuses the existing preprocessing operator unchanged: removal of instrument-edge bands with systematically zero values, followed by standard normal variate (SNV) normalization [9]. For Raman, the preprocessing chain — extended, not rewritten, from the same software contract — comprises: (i) cosmic-ray removal via a modified Z-score computed on the first derivative [10], restricted to contiguous runs of at most three points so that the steep edges of genuine narrow Raman peaks are not misidentified as spikes; (ii) baseline correction by asymmetric least squares [11], removing the fluorescence background characteristic of Raman measurements; and (iii) SNV normalization.

4.3. Architecture

Let $\theta \in \mathbb{R}^p$ be the vector of gate parameters, one per spectral variable. The gate is obtained by applying the logistic sigmoid elementwise,

$$g_i = \sigma(\theta_i) = \frac{1}{1 + e^{-\theta_i}} \in (0, 1), \qquad i = 1, \ldots, p, \tag{1}$$

and the gated spectrum $x \odot g$ (elementwise product) is passed to a shallow MLP with one hidden layer of 32 units, ReLU activation, and dropout. Both branches use this identical model, `PGSGv2Model`. The gate parameters θ and

the MLP weights are trained jointly, end to end, by backpropagation (Adam optimizer [12], weight decay 10^{-4}). Early stopping monitors the loss on a validation subset of 20% of the training partition, selected systematically after sorting the training samples by target value so that it spans the target range; this selection is deterministic and therefore identical across training seeds, so that seed-to-seed variation reflects only the initialization of the MLP weights and the order of mini-batches. The test partition is never used for training, early stopping, or model selection. Before training, θ is initialized from a prior $s \in (0,1)^p$ through the logit transform, $\theta_0 = \mathrm{logit}(s)$, so that g starts exactly at s. No architectural component differs between modalities; this identity constitutes the condition under which H1 is tested.

4.4. Priors

For NIR, the prior is a fixed step function over three literature-derived spectral regions relevant to dry matter content — 680–700 nm (chlorophyll), 900–1000 nm (second C–H overtone), and 1100–1200 nm (first C–H overtone) [5] — independent of the training data. For Raman, the prior is a mixture of Gaussian components centered on literature-reported glucose Raman bands, at two confidence tiers: three primary bands (911, 1060, and 1125 cm^{-1}) supported by multiple independent sources [13, 14], and a secondary set of sixteen bands specific to bioprocess culture-medium monitoring, from a single source [15]. In both modalities the prior is constructed solely from published band assignments, never from the experiment's own training data — a design choice made to avoid the circularity that would result from deriving the prior and the target from the same sample.

4.5. Gate topology descriptors

Let $g \in (0,1)^p$ denote the learned gate. Because the gate is an independent sigmoid per variable, $\sum_i g_i \neq 1$ in general, unlike a softmax gate; entropy is therefore computed on the normalized gate. The three descriptors

are defined as

$$H(g) = -\sum_{i=1}^{p} \hat{g}_i \log \hat{g}_i, \qquad \hat{g}_i = \frac{g_i}{\sum_j g_j} \tag{2}$$

$$\mathrm{TV}(g) = \frac{1}{p-1} \sum_{i=1}^{p-1} |g_i - g_{i+1}| \tag{3}$$

$$S(g) = \frac{\sqrt{p} - \|g\|_1/\|g\|_2}{\sqrt{p}-1} \in [0,1] \tag{4}$$

where Eq. (4) is the sparsity index of Hoyer [16]; $S(g) = 0$ corresponds to a uniformly dense gate and $S(g) = 1$ to a one-hot gate. Cross-modality comparison of $\mathrm{TV}(g)$ requires an additional normalization by physical band density, described in Section 6.2.

4.6. Experimental protocol

All experiments use a within-dataset (intra-domain) protocol with randomly drawn 80/20 train/test splits. The gate analyses (H2–H5) use a single reference split (random seed 42) that, as in the submitted version, does not keep replicate spectra of the same sample together; the predictive comparison (H1) uses ten additional splits grouped by sample, described in Section 4.7. Cross-season or cross-instrument protocols were deliberately avoided, because the distribution shift they introduce would confound the modality comparison with domain shift.

4.7. PLS baseline and H1 protocol

The number of PLS latent variables H was selected on the training partition only, by 5-fold cross-validation over $H = 1, \ldots, 120$, with folds formed by whole samples (fruit or mixture), using two rules: the minimum of the mean cross-validated squared error (PLS-cvmin) and the smallest H whose error lies within one standard error of that minimum (PLS-1se). PLS-1se is the predeclared reference for H1. PLS with a fixed $H = 10$ (PLS-10), the baseline of the submitted version of this manuscript, is reported for comparison. H1 was evaluated over ten random 80/20 splits of whole samples ($r = 0, \ldots, 9$), so that no fruit or mixture contributes spectra to both partitions; in split r, all models are fitted on the same training partition and `PGSGv2Model` uses random seed r, so that comparisons are paired. H1 is considered supported in a modality if the mean paired difference $\Delta R^2 = R^2_{\mathrm{PGSGv2}} - R^2_{\mathrm{PLS\text{-}1se}}$ is positive and a two-sided Wilcoxon signed-rank test gives $p < 0.05$.

5. Notes on model identity and baseline calibration

During execution of this study, an earlier and never-published PGSG implementation — a softmax gate over a PLS regressor optimized by finite-difference gradients — was inadvertently substituted for `PGSGv2Model` in an initial round of Raman experiments. This earlier implementation had never been used to produce any previously published result. Under it, gate training on Raman terminated at the very first epoch, with no improvement over the prior-initialized gate at any subsequent step, and $\Delta R^2_{\mathrm{PLS}} \approx 0$, leading to an initial and incorrect conclusion that H1 was not supported, tentatively attributed to gate over-parameterization. Once the discrepancy was identified and the correct architecture substituted, the opposite result was obtained (Section 6.1). We report this explicitly, together with the diagnostic evidence for both versions, as part of a transparent scientific record; the associated code, data, and commit history are publicly available (Section 8).

A second correction concerns the PLS baseline. In the submitted version of this manuscript, PLS used a fixed number of latent variables ($H = 10$), a value inherited from an earlier dataset and not re-selected for the present ones. During revision, H was selected by cross-validation on the training data (Section 4.7). In NIR, the one-standard-error rule selected H between 46 and 72 (minimum-error rule: 78–92), and the calibrated baseline reversed the conclusion for H1 in that modality; in Raman, H between 4 and 8 was selected and the conclusion was unchanged (Section 6.1). A third correction concerns replicate spectra: the submitted version split spectra at random, so that the two spectra of a fruit or mixture could fall into both partitions. The revised H1 analysis splits whole samples. Relative to random splits, grouping lowered the mean test R^2 of every model by at most 0.017 and changed no conclusion. Both the fixed-H and the calibrated results are reported, and the revision scripts are in the public repository.

6. Results

6.1. H1: predictive performance against PLS

Table 2 and Fig. 1 report test-set R^2 over ten paired splits grouped by sample. In NIR, `PGSGv2Model` ($R^2 = 0.794 \pm 0.024$) is outperformed by calibrated PLS (0.902 ± 0.016 for PLS-1se, 0.907 ± 0.016 for PLS-cvmin): the paired difference is $\Delta R^2 = -0.108 \pm 0.031$, negative in all ten splits (Wilcoxon $p = 0.002$). Against PLS-10 the same model appears to outperform PLS by

$\Delta R^2 = +0.231 \pm 0.035$, which reproduces the advantage reported in the submitted version (+0.244 on the reference split); that advantage is therefore attributable entirely to an under-parameterized baseline, since the cross-validation error of PLS keeps decreasing well beyond $H = 10$. In Raman, `PGSGv2Model` ($R^2 = 0.932 \pm 0.008$) outperforms calibrated PLS (0.896 ± 0.011 for PLS-1se, 0.902 ± 0.011 for PLS-cvmin): $\Delta R^2 = +0.036 \pm 0.007$, positive in all ten splits (Wilcoxon $p = 0.002$). Here the fixed baseline was already close to the calibrated one (H of 4–8 selected), and the advantage is essentially that of the submitted version. **H1 is not supported in NIR and is supported in Raman.**

On the reference split used for the gate analyses, training proceeds well past the first epoch in both modalities — the best validation epoch is 79 (of up to 500, patience 30) in Raman and 469 in NIR — while the learned gate remains close to its initialization in correlation, $\rho = 0.978$ (Raman) and $\rho = 0.998$ (NIR). These two facts are compatible: the prior supplies a strong structure, so a substantive movement of the gate still leaves the correlation with the starting point high. Section 6.3 quantifies that movement directly.

Table 2: H1: test-set R^2 (mean ± SD over ten paired random splits) and paired difference with respect to `PGSGv2Model`. Splits keep all spectra of a fruit (NIR) or mixture (Raman) in the same partition. H selected by the one-standard-error rule: 46–72 (NIR), 4–5 (Raman); minimum-error rule: 78–92 (NIR), 7–8 (Raman).

	NIR		Raman	
Model	R^2	ΔR^2	R^2	ΔR^2
PLS-10 ($H = 10$)	0.563 ± 0.043	$+0.231 \pm 0.035$	0.900 ± 0.010	$+0.032 \pm 0.004$
PLS-cvmin	0.907 ± 0.016	-0.113 ± 0.029	0.902 ± 0.011	$+0.030 \pm 0.005$
PLS-1se	0.902 ± 0.016	-0.108 ± 0.031	0.896 ± 0.011	$+0.036 \pm 0.007$
`PGSGv2Model`	0.794 ± 0.024	—	0.932 ± 0.008	—

6.2. H2 and H3: gate topology across modalities

H2 (sparsity) is supported and robust to the functional form of the prior. With each modality's original prior form, Hoyer sparsity is $S_{\mathrm{NIR}} = 0.3176$ versus $S_{\mathrm{Raman}} = 0.4150$ (Fig. 2a). To rule out an encoding artifact — the NIR prior is a step function, whereas the Raman prior is a Gaussian mixture — the NIR prior was recoded as Gaussian bumps at the same literature centers and relative weights, matching the Raman encoding.

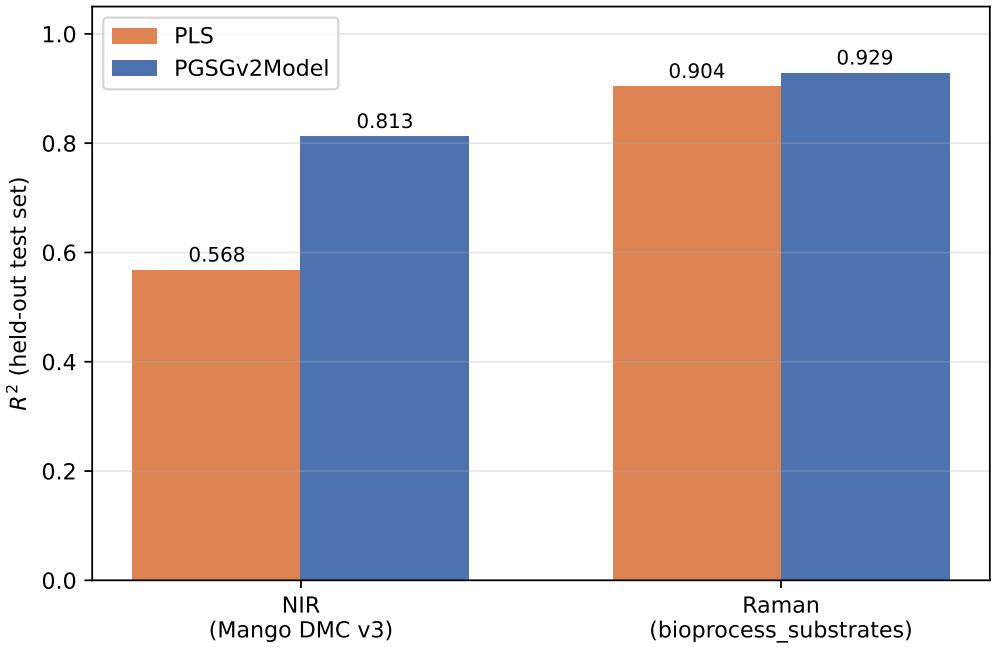


Figure 1: H1: test-set R^2 over ten paired splits, grouped by sample, for PLS with fixed $H = 10$, PLS with cross-validated H (cvmin and one-standard-error rules), and `PGSGv2Model`. Bars: mean; error bars: SD; points: individual splits.

The difference persists and widens ($S_{\mathrm{NIR}} = 0.2345$ versus $S_{\mathrm{Raman}} = 0.4150$), indicating that the effect is not attributable to prior-encoding style.

H3 (smoothness) requires a methodological correction. A first comparison of $\mathrm{TV}(g)$ by raw variable index yielded the result opposite to that predicted: $\mathrm{TV}_{\mathrm{NIR}} = 0.0167 > \mathrm{TV}_{\mathrm{Raman}} = 0.0076$ (Fig. 2b). This reversal persisted under the Gaussian-prior ablation and is therefore not a prior-encoding artifact. Its cause is a difference in spectral sampling density: the NIR axis ($p = 281$ variables over $\approx 26{,}755$ cm^{-1}, ≈ 95 cm^{-1} per variable) is approximately sixty times sparser, in wavenumber terms, than the Raman axis ($p = 1{,}870$ variables over $\approx 2{,}995$ cm^{-1}, ≈ 1.6 cm^{-1} per variable). Total variation computed per variable index conflates the number of variables separating two informative regions with the smoothness of the transition per unit of physical spectrum, and is therefore an unsuitable metric for comparison across modalities with such disparate variable densities.

Recomputing smoothness per unit of physical bandwidth — converting the NIR axis from nanometers to wavenumbers via $\tilde{\nu} = 10^7/\lambda$ — reverses the direction, now matching the hypothesis: $\mathrm{TV}_{\mathrm{physical,NIR}} = 3.69 \times 10^{-4} \ll \mathrm{TV}_{\mathrm{physical,Raman}} = 3.80 \times 10^{-3}$ (Fig. 2c). This is corroborated by an independent check, unrelated to any gate or prior: the raw preprocessed spectrum is approximately 2.3-fold rougher (point-to-point variation) in Raman than in NIR (0.2156 versus 0.0926). Two independent measures agree once the density confound is removed. **H3 is supported**, provided smoothness is measured per unit of physical spectrum rather than per raw variable index; we note that the raw-index comparison is unreliable for cross-modality use in general and should be avoided in future work of this kind.

6.3. H4: value of a physically motivated prior

Across ten training seeds per condition, predictive performance does not differ between literature-informed and uninformed (zero) initialization beyond seed-to-seed variability in either modality (Table 3); the differences in mean R^2 (0.0050 NIR, 0.0008 Raman) are smaller than one standard deviation across seeds: NIR, $R^2_{\mathrm{lit}} = 0.7933 \pm 0.0080$ versus $R^2_{\mathrm{rand}} = 0.7983 \pm 0.0077$; Raman, $R^2_{\mathrm{lit}} = 0.9302 \pm 0.0015$ versus $R^2_{\mathrm{rand}} = 0.9310 \pm 0.0014$. Gate stability across seeds, in contrast, differs sharply and consistently in favor of the informed prior (Fig. 3): pairwise gate correlation $\rho = 0.9995$ (NIR, literature prior) versus 0.8653 (NIR, uninformed), and 0.9976 (Raman, literature prior) versus 0.8931 (Raman, uninformed); pairwise Jaccard similarity of the top-weighted decile of variables, 0.9418 (NIR, literature prior) versus 0.6002

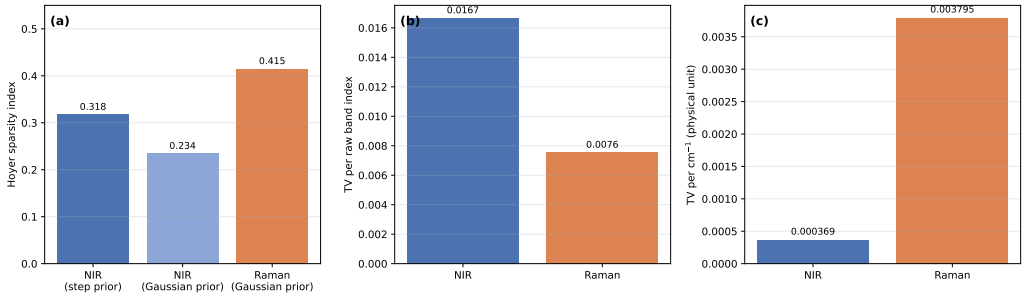


Figure 2: Gate topology across modalities. (a) Hoyer sparsity: Raman more sparse than NIR under both prior encodings. (b) Smoothness by raw variable index: a misleading comparison, dominated by the difference in spectral sampling density between modalities. (c) Smoothness normalized by physical unit (cm^{-1}): the correct comparison, in agreement with raw-spectrum roughness.

(NIR, uninformed), and 0.9011 (Raman, literature prior) versus 0.6545 (Raman, uninformed).

Table 3: H4: performance and gate stability, literature prior versus uninformed initialization (10 seeds).

	NIR		Raman	
	Literature prior	Uninformed	Literature prior	Uninformed
R^2 (mean $\pm$ SD)	0.7933 ± 0.0080	0.7983 ± 0.0077	0.9302 ± 0.0015	0.9310 ± 0.0014
ρ (pairwise)	0.9995	0.8653	0.9976	0.8931
Jaccard (pairwise)	0.9418	0.6002	0.9011	0.6545

H4 is supported for gate stability; the accuracy component of the hypothesis ("equal to or better than") is not contradicted, but no superiority was observed, and equivalence was not formally tested (e.g., by two one-sided tests with a pre-declared margin), in either modality.

6.4. H5: structure–physics correlation

H5, as originally stated across modalities, is not testable as a statistical correlation with only two modalities ($n = 2$). We reformulate it as a within-modality, band-by-band test in Raman, correlating literature-reported linewidths for twenty glucose Raman bands [15] against three descriptors extracted locally around each band from the trained gate, within an adaptive window capped at 45% of the distance to the nearest neighboring literature

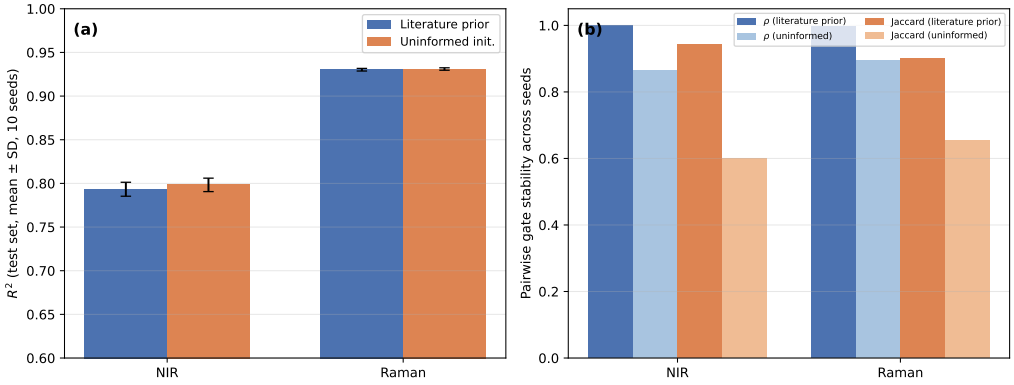


Figure 3: H4: literature-informed prior versus uninformed initialization, 10 training seeds. (a) Predictive performance does not differ beyond seed-to-seed variability. (b) Gate stability across seeds is substantially higher with the literature-informed prior, in both modalities.

band, to avoid cross-contamination between closely spaced bands such as the 1060/1073 cm^{-1} pair. NIR is excluded from this quantitative test: it has only three literature bands, and its prior already encodes their exact literature widths by construction, which would render any correlation circular.

None of the three local descriptors yields a robust, correctly signed correlation (Fig. 4, $n = 20$): local gate full width at half maximum, $r = +0.254$ ($p = 0.279$); local smoothness, $r = +0.048$ ($p = 0.840$); local entropy, computed raw, $r = +0.407$ ($p = 0.075$, marginal, correct direction), and normalized by window size ($\log n$), $r = -0.636$ ($p = 0.003$, significant but in the direction opposite to that predicted). Diagnosis traced the entropy instability to the adaptive window itself: narrow windows, required near closely spaced bands, sample predominantly the flat top of a Gaussian bump, mechanically inflating normalized entropy toward its ceiling regardless of true local structure, whereas wider windows capture more of the true rise and fall, pulling the metric down. A synthetic control with a known, built-in correlation (gate bump width proportional to literature width by construction) confirmed that the extraction method recovers the expected correlation when it exists ($r = 0.73$, 0.57, and -0.86 for the three descriptors, all $p < 0.01$), indicating that the null result on real data is informative rather than a failure of the extraction procedure. **H5 is not supported.** Further refinement of the windowing scheme was not pursued, to avoid the risk of adjusting a method until it yields the expected significance.

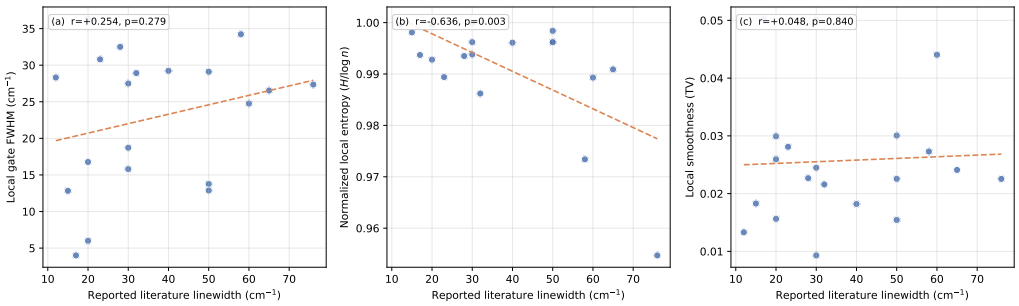


Figure 4: H5: literature-reported linewidth versus local gate descriptors, Raman, $n = 20$ bands. Dashed lines show ordinary least-squares fits. None of the three descriptors shows a robust, correctly signed correlation.

7. Discussion

Table 4 summarizes the outcome of the five hypotheses.

Taken together, the results separate two questions that the submitted version of this work had conflated: whether the gated model predicts better than PLS, and whether its gate carries interpretable, physically consistent structure. On the first question the answer depends on the modality, and in the direction opposite to the one originally reported. In NIR, a PLS model with some seventy latent variables captures the dry-matter signal of this heterogeneous, multi-cultivar season better than the shallow MLP; the large NIR advantage of the submitted version was an artifact of an under-parameterized baseline. In Raman, PLS saturates at four to eight latent variables and the gated MLP improves on it consistently. Because the learned gate stays close to its initialization ($\rho \geq 0.98$) and an uninformed initialization reaches the same accuracy (H4), this gain cannot be credited to the prior; it is more plausibly a property of the non-linear regressor, for instance its ability to represent interactions among the eight co-varying substrates of the bioprocess mixtures, which we did not test directly. The contrast between fixed-H and calibrated comparisons is itself a methodological result: predictive claims for neural spectral models should be made only against a linear baseline whose complexity is selected by cross-validation, with training, validation, and test partitions that keep replicate spectra of a sample together. On the second question the answer is positive: gate topology differs between modalities in the direction predicted by their physics (H2, H3), provided the comparison is made per unit of physical spectrum.

Table 4: Summary of hypothesis tests.

Hyp.	Outcome	Key evidence
H1	Not supported (NIR); supported (Raman)	Paired ΔR^2 vs. PLS-1se: -0.108 ± 0.031 (NIR, 0/10 splits positive); $+0.036 \pm 0.007$ (Raman, 10/10)
H2	Supported, robust	Hoyer sparsity: Raman 0.415 ¿ NIR 0.235–0.318 (two prior encodings)
H3	Supported, with correction	Smoothness per cm^{-1}: NIR $3.69 \times 10^{-4} \ll$ Raman 3.80×10^{-3}; consistent with raw-spectrum roughness
H4	Supported (stability); no performance difference detected	Pairwise gate ρ: 0.998–0.9995 (prior) versus 0.86–0.89 (uninformed), both modalities
H5	Not supported	No local descriptor correlates robustly with real per-band linewidth ($n = 20$, Raman)

H4 refines this picture further: the value of the prior lies not in predictive accuracy — indistinguishable, within seed-to-seed variability, with or without it — but in the reproducibility of the learned attention pattern across independent training runs. A gate that points to different variables on every run is not a trustworthy explanation, even when its predictions are accurate; a physically motivated prior appears to function principally as an anchor for interpretability rather than as a performance lever.

H5, the single negative result, probes a stronger and more granular claim than H2/H3: not merely that aggregate gate topology differs between modalities in a manner consistent with physics (established by H2/H3), but that local gate structure, band by band, tracks the specific linewidth of each vibrational transition. With twenty Raman bands, none of the three descriptors tested produced a robust, replicable correlation, and the entropy descriptor proved unstable enough to reverse sign under a defensible alternative normalization. We interpret this as a resolution limit of the current descriptors rather than evidence that the gate is physically uninformative at a local level: gate topology appears to capture between-modality structural differences without necessarily resolving fine within-modality variation at the level of individual bands, at least with the windowing scheme used here.

7.1. Limitations

Five limitations apply across the study. First, the research process included two consequential corrections: an unpublished, superseded PGSG implementation was inadvertently used for an initial round of Raman experiments, and the PLS baseline of the submitted version used a fixed number of latent variables; each reversed the conclusion for H1 in at least one modality at some stage (Section 5), and both are documented in the public repository. Second, cross-modality topology comparisons require careful normalization by physical band density (Section 6.2); future extensions of this line of work, including within the same research program, should adopt the same caution. Third, H4 was evaluated with ten training seeds per condition; a larger number would strengthen the stability estimate, although the observed effect is already large and consistent across both modalities. Fourth, the predictive comparison is restricted to PLS; comparisons with other attention-based architectures and with sparse linear models (Elastic Net, sparse PLS) were not performed and would be needed to position the model's accuracy more broadly. Fifth, only the H1 analysis uses sample-grouped splits: the gate analyses (H2–H5) use the original random reference split, in which replicate spectra of a sample can fall into both partitions, and the early-stopping validation subset of `PGSGv2Model` is selected by target value within the training partition, without regard to sample grouping. H4 compares two initializations on the same split, so its conclusion is unaffected, but its absolute R^2 values are slightly optimistic.

8. Conclusion

Applied without architectural modification to NIR and Raman spectroscopy, Prior-Guided Spectral Gating outperforms a PLS baseline with cross-validated complexity in Raman but not in NIR, the reverse of the pattern obtained with a fixed number of PLS components; the NIR advantage reported in the submitted version of this work arose from an underparameterized baseline. The learned gate nevertheless exhibits topological differences between the two modalities that are consistent with their physical structure once compared on a physically meaningful basis, and a literature-informed prior makes the gate substantially more reproducible across training runs without changing accuracy. A finer, band-level test of structure–physics correspondence did not succeed with the descriptors used here. Because an uninformed initialization reaches the same accuracy, the Raman gain belongs

to the non-linear regressor rather than to the prior. For subsequent stages of the research program (multi-target regression, uncertainty quantification), the case for spectral gating should rest on the stability and physical interpretability of the gate, evaluated against calibrated linear baselines and with sample-grouped validation, rather than on an assumed predictive gain.

Data and code availability

All code, experiment scripts, saved model outputs, and the commit history documenting the model-identity correction described in Section 5 are available at `https://github.com/clarimar/pgsg2-nir-raman-gating`. The repository includes a verified snapshot of the model and baseline implementations (`vendor/pgsg_1`, with SHA-256 checksums), so that all experiments can be run from it alone. The Raman data were obtained through the `raman-data` package from the Hugging Face dataset `chlange/SubstrateMixRaman`, revision `887cf66d8cb25b2b953ee0a36889812f684de03e`.

Acknowledgments

The authors gratefully acknowledge the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Brazil, for covering the Article Processing Charge of this publication under the CAPES–Elsevier Read and Publish Agreement (2026–2028). This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

[1] C. J. Coelho, et al., Prior-guided spectral gating for interpretable deep learning in near-infrared spectroscopy, Chemometrics and Intelligent Laboratory SystemsAccepted for publication (2026).

[2] J. Workman, Jr., L. Weyer, Practical Guide and Spectral Atlas for Interpretive Near-Infrared Spectroscopy, 2nd Edition, CRC Press, 2012.

[3] E. Smith, G. Dent, Modern Raman Spectroscopy: A Practical Approach, 2nd Edition, Wiley, 2019.

[4] N. T. Anderson, K. B. Walsh, P. P. Subedi, C. H. Hayes, Achieving robustness across season, location and cultivar for a NIRS model for

intact mango fruit dry matter content, Postharvest Biology and Technology 168 (2020) 111202.

[5] S. Saranwong, J. Sornsrivichai, S. Kawano, Prediction of ripe-stage eating quality of mango fruit from its harvest quality measured nondestructively by near infrared spectroscopy, Postharvest Biology and Technology 31 (2004) 253–261.

[6] C. Lange, et al., Deep learning for Raman spectroscopy: Benchmarking models for upstream bioprocess monitoring, Measurement 258 (2026) 118884.

[7] M. Koddenbrock, C. Lange, R. Legner, M. Jaeger, M. Kögler, et al., RamanBench: A large-scale benchmark for machine learning on Raman spectroscopy, arXiv preprint arXiv:2605.02003 (2026).

[8] A. Echtermeyer, C. Marks, A. Mitsos, J. Viell, Inline Raman spectroscopy and indirect hard modeling for concentration monitoring of dissociated acid species, Applied Spectroscopy 75 (5) (2021) 506–519.

[9] R. J. Barnes, M. S. Dhanoa, S. J. Lister, Standard normal variate transformation and de-trending of near-infrared diffuse reflectance spectra, Applied Spectroscopy 43 (5) (1989) 772–777.

[10] D. A. Whitaker, K. Hayes, A simple algorithm for despiking Raman spectra, Chemometrics and Intelligent Laboratory Systems 179 (2018) 82–84.

[11] P. H. C. Eilers, H. F. M. Boelens, Baseline correction with asymmetric least squares smoothing, Tech. rep., Leiden University Medical Centre (2005).

[12] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: Proceedings of the 3rd International Conference on Learning Representations (ICLR), 2015.

[13] N. Lyu, et al., In vivo blood glucose quantification using Raman spectroscopy, PLOS ONE 7 (11) (2012) e48127.

[14] J. W. Kang, et al., Direct observation of glucose fingerprint using in vivo Raman spectroscopy, Science Advances 6 (4) (2020) eaay5206.

[15] US Patent, Use of Raman spectroscopy to monitor culture medium (2024).

[16] P. O. Hoyer, Non-negative matrix factorization with sparseness constraints, Journal of Machine Learning Research 5 (2004) 1457–1469.